NUMERISCHE MATHEMATIK I | NEWTON VERFAHREN

JONAS BRESCH

1. Review – The Newton Method

Let $n \in \mathbb{N}$ and $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. We assume there exists a solution $x^* \in \mathbb{R}^n$ to the nonlinear equation

$$F(x^*) = 0. \tag{1.1}$$

Our goal is to find such an x^*. A first-order Taylor expansion of F at a point $x_0 \in \mathbb{R}^n$ yields

$$F(x^*) = F(x_0) + J_F(x_0)\,(x^* - x_0) + \mathcal{O}(\|x^* - x_0\|_2^2),$$

where $J_F(x)$ denotes the Jacobian matrix of F at x. If $J_F(x_0)$ is invertible and $\|x^* - x_0\|_2$ is small, we can solve the approximation

$$0 = F(x_0) + J_F(x_0)\,(x^* - x_0) \tag{1.2}$$

for x^* and obtain

$$x^* = x_0 - J_F(x_0)^{-1} F(x_0).$$

This motivates the iterative method:

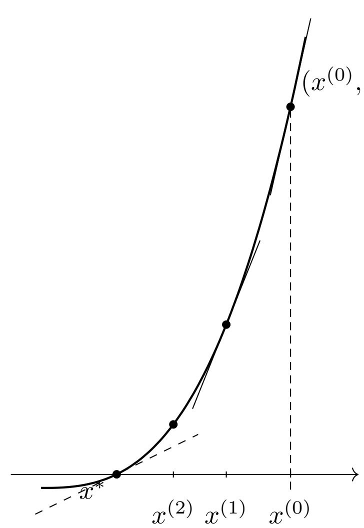
- Choose starting value $x^{(0)} \in \mathbb{R}^n$.
- For $k = 0, 1, 2, \ldots$ set

$$x^{(k+1)} := x^{(k)} - J_F\big(x^{(k)}\big)^{-1} F\big(x^{(k)}\big), \tag{1.3}$$

  provided $J_F(x^{(k)})$ is invertible.

Since inverting matrices can be numerically unfavorable and expensive, the iteration is written in two steps:

Date: 7. Januar 2026.



1. Solve the linear system of equations
$$J_F\big(x^{(k)}\big)\, d^{(k)} = -F\big(x^{(k)}\big)$$
for $d^{(k)}$.
2. Set $x^{(k+1)} = x^{(k)} + d^{(k)}$.

The method is terminated as soon as one of the following stopping criteria is met:

a. Iteration difference: $\|x^{(k)} - x^{(k-1)}\|_2 < \delta$,
b. Function value: $\|F(x^{(k)})\|_2 < \varepsilon$,
c. Maximum number of iterations: $k > K$,

with suitable parameters $\delta, \varepsilon > 0$ and $K \in \mathbb{N}$.

In the one-dimensional case ($n = 1$), the iteration corresponds to finding the root of the linear Taylor approximation at the point $x^{(k)}$.

2. Applications of the Newton Method

2.1. The d-th Root of Unity. The Newton method can also be applied to complex differentiable functions $f : \mathbb{C} \to \mathbb{C}$. To determine a d-th root of unity, one looks for roots of
$$z^d - 1 = 0.$$
The Newton iteration for this function is
$$\begin{aligned} z^{(k+1)} &= z^{(k)} - \frac{(z^{(k)})^d - 1}{d\,(z^{(k)})^{d-1}} \\ &= \frac{1}{d}\Big((d-1)\, z^{(k)} + \frac{1}{(z^{(k)})^{d-1}}\Big). \end{aligned}$$

2.2. Nonlinear Optimization. The Newton method can be used to determine a minimum $x^* \in \mathbb{R}^n$ of a twice continuously differentiable function $G : \mathbb{R}^n \to \mathbb{R}$. We apply the Newton method to the gradient $F := \nabla G$. Using the second-order Taylor expansion
$$G(x) \approx G\big(x^{(k)}\big) + \nabla G\big(x^{(k)}\big)^T (x - x^{(k)}) + \tfrac{1}{2}(x - x^{(k)})^T H_G\big(x^{(k)}\big)(x - x^{(k)}),$$
we set the derivative with respect to x to zero and obtain
$$\nabla G\big(x^{(k)}\big) + H_G\big(x^{(k)}\big)(x - x^{(k)}) = 0,$$
from which the iteration
$$x^{(k+1)} = x^{(k)} - H_G\big(x^{(k)}\big)^{-1}\, \nabla G\big(x^{(k)}\big) \tag{2.1}$$
follows, provided the Hessian matrix $H_G(x^{(k)})$ is positive definite. Note that the Newton method generally only converges to stationary points (minimum, maximum, saddle point).

Summary

- Newton solves nonlinear systems of equations via local linearization.
- Numerically implemented by solving a linear system of equations in each step.
- Applications: complex roots (e.g., d-th roots of unity), optimization (applied to ∇G).

3. Numerical Examples

4. Introduction

In this session, we explore the implementation of the Newton method for finding roots of functions $F : \mathbb{R}^n \to \mathbb{R}^n$ and minimizing scalar functions[cite: 6]. We further examine the convergence behavior of the method in the complex plane, leading to the generation of fractals.

5. The Newton Method

Let $n \in \mathbb{N}$ and $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable. We aim to find a solution $x^* \in \mathbb{R}^n$ to the nonlinear equation $F(x^*) = 0$.

The iterative formula is derived from the first-order Taylor approximation $0 = F(x^{(k)}) + J_F(x^{(k)})(x^{(k+1)} - x^{(k)})$, leading to:

$$x^{(k+1)} = x^{(k)} - J_F\big(x^{(k)}\big)^{-1} F\big(x^{(k)}\big) \tag{5.1}$$

where J_F is the Jacobian matrix. Numerically, this is implemented by solving a linear system for the step $d^{(k)}$:

(1) Solve the linear system: $J_F\big(x^{(k)}\big)\, d^{(k)} = -F\big(x^{(k)}\big)$
(2) Update the solution: $x^{(k+1)} = x^{(k)} + d^{(k)}$

Exercise 1: Implementation. We implement a generic function `newton(F, dF, x0, delta, epsilon, maxIter)`[cite: 7].

- **Stopping Criteria:** $|x^{(k+1)} - x^{(k)}| < \delta$ or $\|F(x^{(k)})\| < \epsilon$[cite: 8].
- **Defaults:** $\delta = \epsilon = 10^{-4}$, maxIter = 100.

Algorithm 1 Newton's Method for $F(x) = 0$

Require: Function F, Jacobian J_F, initial guess $x^{(0)}$, tolerances δ, ε, max iterations K.

$x \leftarrow x^{(0)}$
for $k = 0$ to $K - 1$ **do**
  $F_k \leftarrow F(x)$
  if $\|F_k\|_2 < \varepsilon$ **then return** x ▷ Stopping criterion b: Function value
  end if
  $J_k \leftarrow J_F(x)$
  Solve $J_k d = -F_k$ for the step d
  $x_{\text{new}} \leftarrow x + d$
  if $\|x_{\text{new}} - x\|_2 < \delta$ **then return** x_{new} ▷ Stopping criterion a: Iteration difference
  end if
  $x \leftarrow x_{\text{new}}$
end forreturn x ▷ Stopping criterion c: Max iterations reached

6. Applications in $\mathbb{R}$ and $\mathbb{R}^2$

Exercise 2: Scalar Root Finding. We find the roots of the function $f : \mathbb{R} \to \mathbb{R}$:

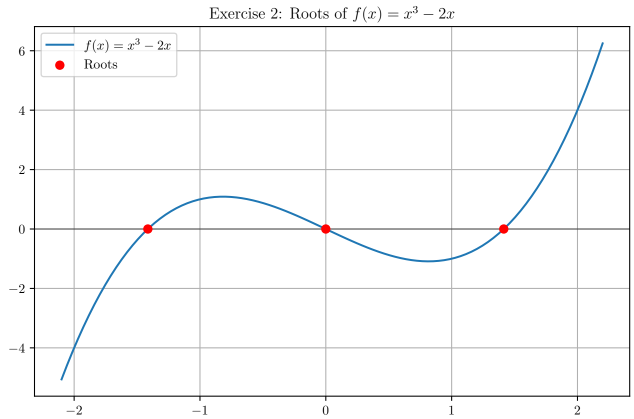
$$f(x) = x^3 - 2x \tag{6.1}$$

The derivative $f'(x)$ is required for the Newton iteration:

$$f'(x) = \frac{d}{dx}(x^3 - 2x) = 3x^2 - 2$$

The iterative step is $x^{(k+1)} = x^{(k)} - \frac{f(x^{(k)})}{f'(x^{(k)})}$. Using the prescribed starting values:

- $x^{(0)} = 0.1$: Converges to $x^* = 0$.
- $x^{(0)} = 2$: Converges to $x^* = \sqrt{2} \approx 1.414214$.
- $x^{(0)} = -2$: Converges to $x^* = -\sqrt{2} \approx -1.414214$.



Exercise 3: System of Equations. We solve for the roots of $F : \mathbb{R}^2 \to \mathbb{R}^2$:

$$F(x) = \begin{pmatrix} x_1^2 + x_2^2 - 6x_1 \\ \frac{3}{4}e^{-x_1} - x_2 \end{pmatrix} \tag{6.2}$$

Starting at $x^{(0)} = (0.08, 0.7)^T$.

7. Newton Fractals in $\mathbb{C}$

We solve the system $F(x) = 0$, where $x = (x_1, x_2)^T$:

$$F(x) = \begin{pmatrix} F_1(x) \\ F_2(x) \end{pmatrix} = \begin{pmatrix} x_1^2 + x_2^2 - 6x_1 \\ 0.75e^{-x_1} - x_2 \end{pmatrix} \tag{7.1}$$

The Jacobian matrix $J_F(x)$ is calculated as:

$$J_F(x) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \frac{\partial F_1}{\partial x_2} \\ \frac{\partial F_2}{\partial x_1} & \frac{\partial F_2}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 2x_1 - 6 & 2x_2 \\ -0.75e^{-x_1} & -1 \end{pmatrix} \tag{7.2}$$

Starting at $x^{(0)} = (0.08, 0.7)^T$, the Newton method converges to the root:

$$x^* \approx (0.080833, 0.691761)^T$$

Exercise 4: The Cubic Unity Roots. The Newton iteration for the general problem $f(z) = z^d - 1 = 0$ is:

$$z^{(k+1)} = z^{(k)} - \frac{f(z^{(k)})}{f'(z^{(k)})} = z^{(k)} - \frac{(z^{(k)})^d - 1}{d(z^{(k)})^{d-1}}$$

Setting $d = 3$, the specific iteration used to generate the fractal is:

$$z^{(k+1)} = \frac{2}{3}z^{(k)} + \frac{1}{3(z^{(k)})^2}$$

The complex plane $B_\infty = [-1, 1] \times [-1, 1]$ is analyzed, coloring each point based on which of the three roots of unity it converges to.

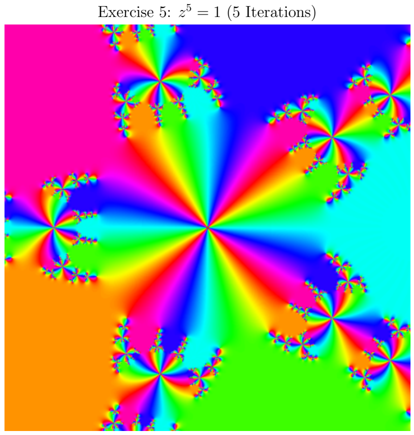
Exercise 5: Phase Plots of Higher Order. Setting $d = 5$, the iteration is:

$$z^{(k+1)} = \frac{4}{5}z^{(k)} + \frac{1}{5(z^{(k)})^4}$$

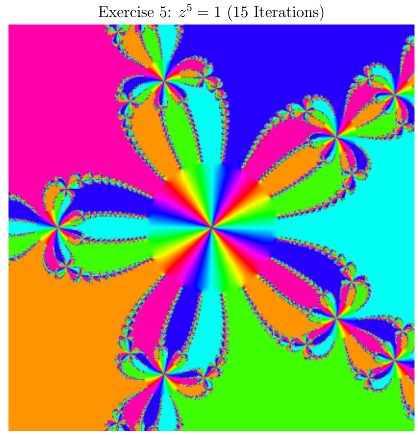
The image visualizes the argument (phase) of the resulting complex number $z^{(15)}$ for two different numbers of iterations, highlighting the complex boundary behavior.

Exercise 4: Newton Fractal $z^3 = 1$

Exercise 5: $z^5 = 1$ (5 Iterations)



Exercise 5: $z^5 = 1$ (15 Iterations)



8. Optimization via Newton

Exercise 6: Minimization. To minimize a function $G : \mathbb{R}^n \to \mathbb{R}$, we seek roots of its gradient $F(x) = \nabla G(x) = 0$. The Newton update uses the Hessian $H_G(x)$ as the Jacobian of $F(x)$. We minimize:

$$G(x) = (x_1 + 1)^2 + (x_2 - 1)^4 \tag{8.1}$$

The required gradient is:

$$\nabla G(x) = F(x) = \begin{pmatrix} 2(x_1 + 1) \\ 4(x_2 - 1)^3 \end{pmatrix}$$

The required Hessian (Jacobian of ∇G) is:

$$H_G(x) = J_F(x) = \begin{pmatrix} 2 & 0 \\ 0 & 12(x_2 - 1)^2 \end{pmatrix}$$

Starting at $x^{(0)} = (-1.1, 1.1)^T$, the method finds the minimum at:

$$x^* = (-1.0, 1.0)^T$$

9. The Mandelbrot Set

Exercise 7: Divergence Test. We analyze the behavior of the iteration:

$$z^{(k+1)} = (z^{(k)})^2 + c \quad \text{with } z^{(0)} = c \tag{9.1}$$

For points c in the domain $M = [-1.5, 0.5] \times [-1, 1]$, we count the number of iterations $k \leq 256$ for which the magnitude $|z^{(k)}| \leq 2$. The count serves as a measure of divergence speed, which is used to color the resulting fractal image.